Phenomena Illustrating the Laws of Motion

SYNOPSIS

This note runs backwards from the others. There is no equation to reach and nothing to derive. You watch four things happen, write down what you notice before you know what to call it, and only then ask which of Newton's laws is doing the work. Two of the four you will be able to settle today. **Two of them you will not** — and knowing exactly why you cannot settle them yet is worth as much as settling them.

BEFORE YOU READ ON — PREDICT

A penny is sitting on a card, and the card is resting across the mouth of a cup. Using **one finger**, and without touching the penny, get the penny into the cup.

Write down the method you would try, and what you expect the penny to do, before you turn the page.

PERFORMANCE EXPECTATIONS

1. **Record** an observation before explaining it — what moved, how it moved, and what changed — without reaching for vocabulary you have not justified.
2. **Name** the object of interest in a demonstration, and **explain** why the choice of object decides which forces appear in the free-body diagram.
3. **Identify** which of Newton's laws accounts for a demonstrated outcome, and **state** what you would see instead if the force in question were absent.
4. **Distinguish** a phenomenon you can account for with the tools you have from one you cannot, and say **which tool is missing**.

How to work a phenomenon

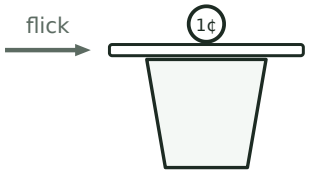
THE FOUR QUESTIONS

object → motion → energy → force

PHENOMENON 1 — A PENNY INTO A CUP

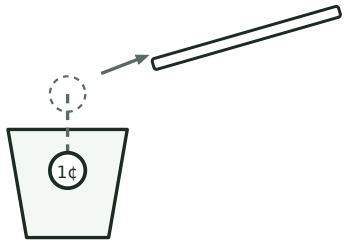
Flick the card

BEFORE



penny on a card, card on the cup

AFTER



the card leaves sideways. the penny does not.

1. Write at least three things you **notice** and three things you **wonder**. No explanations yet.

ASK	YOUR ANSWER
Object of interest What is making the interesting motion?	
The motion Speeding up, slowing down, or no change? Straight or not?	
Energy More or less energy in the object? Where did it come from?	
Force(s) Which force, on which body, and what is it doing?	

2. The only thing touching the penny is the card, and the card is only in contact with it for a few thousandths of a second. What force acts on the penny during that time, and roughly how large an effect can it have?

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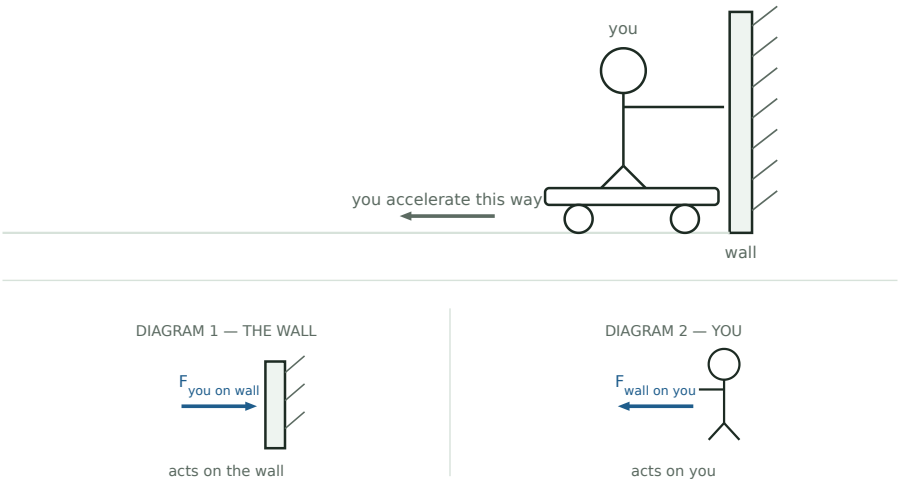
3. Now answer the question the demonstration asks: **why does the penny not go with the card?** Name the law.

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4. Predict: if the card and the penny were glued together, what would happen? What if the card were made of ice and the penny sat in a puddle of oil?

PHENOMENON 2 — PUSHING OFF ON A TROLLEY

Why can you not move without pushing off something?



5. Three things you notice, three things you wonder.

ASK	YOUR ANSWER
Object of interest What is making the interesting motion?	
The motion Speeding up, slowing down, or no change? Straight or not?	
Energy More or less energy in the object? Where did it come from?	
Force(s) Which force, on which body, and what is it doing?	

6. You are the object of interest. **Which force in the picture accelerates you?** Be precise about which body it acts on.

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7. You pushed the wall and the wall pushed back with the same size force. The wall did not move. Explain that without saying the forces were unequal.

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8. Now change the object of interest to **the wall**. Which forces appear in its diagram? Does your answer to the previous question change?

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9. Push harder. What happens to your acceleration, and what does that tell you about the form of the second law?

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10. Predict: on a perfectly frictionless floor with nothing within reach, could you get yourself moving at all? What if you were holding a heavy backpack?

PHENOMENA 3 AND 4 — LEFT OPEN ON PURPOSE

Two you cannot finish yet

OPEN PHENOMENON 3 — BOUNCING A BALL ON A BALL

Why does the small ball bounce so much higher when it is dropped on top of a large bouncy ball than when it is dropped on its own?

- Three things you notice:
- Your best guess, in one sentence:

Why not now. Answering this properly means tracking what each ball carries into the collision and how that is shared out in the instant they touch. Forces alone will not get you there — you need momentum and the impulse of a very short contact, which is a later module. We will come back to this demonstration by name when we have them.

OPEN PHENOMENON 4 — SHOOTING A FALLING TARGET

A target is released at the exact instant the ball is fired straight at it. Why do you hit it every time, no matter how fast you fire or how far away the target is?

- Three things you notice:

- Your best guess, in one sentence:

Why not now. This one is closer than it looks. You already know from Module 2 that the horizontal and vertical parts of a projectile's motion do not interfere with each other. The missing piece is the claim **N4** is about: every object near the Earth falls with the same acceleration, whatever its mass. Put those two together and the result stops being a trick. Hold this one — we settle it in N4.

BACK TO YOUR PREDICTION

You predicted a method for getting the penny into the cup. Did it work? Name the property of the penny that made it work, and say what would have happened had you moved the card slowly.

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